

HENRY B PHILIP

 Henry B Philip | 7034546434 | henrybiju12@gmail.com | Kalady, Kerala

PROFILE

AI Developer with 1.4+ years of experience in Deep Learning, Computer Vision, and LLMs/VLMs. Skilled in training and fine-tuning models, designing agentic workflows, and deploying scalable AI solutions for high-throughput inference of large models using vLLM and Docker. Proficient in building containerized services with FastAPI and workflow orchestration using Celery. Adept at delivering secure, high-performance AI systems with a strong focus on scalability, compliance, and data privacy.

EXPERIENCE

- **AI Developer - PixDynamics Pvt Ltd** (June 2024 - Present)
 - Designed and deployed scalable AI solutions, integrating ML and DL models to automate workflows and enhance decision-making.
 - Contributed across **full project lifecycle: data preparation, model training, testing, optimization, and production deployment.**
 - Built secure, high-performance APIs supporting AI applications and **scalable data extraction pipelines.**
 - Applied NLP and image processing for text classification, information retrieval, document parsing, and **agentic chatbot integration.**
 - Developed **layout-aware document parsing** solutions using **VLMs** to extract structured data from text, figures, and tables.
 - Implemented **security-focused workflows** for sensitive financial and personal data, ensuring privacy and regulatory compliance.
- **AI / ML INTERN - ICT ACADEMY OF KERALA** (May 2023)
 - Developed a predictive model for emotion analysis based on tweet content
- **AI / ML Trainee - TATA ELXSI Ltd. Kozhikode, India** (December 2022)
 - Created ML models for Driver Class Prediction & Cancer Prediction

PROJECTS

- **Multi-Modal AI Chatbot**
 - Engineered an **intelligent chatbot** system capable of handling **multi-modal** inputs (text and documents) with retrieval from **vector databases (Qdrant)**, **SQL databases**, live web sources, and user-uploaded files, using the latest frameworks like **LangChain** and **LangGraph.**
 - Leveraged **GPT-4o**, **Gemini 2.5** and **GPT-OSS** for developing advanced dialogue systems, with RAG pipelines providing factual grounding and reliability.
 - Built **semantic knowledge bases** leveraging **OpenAI embeddings** and **Nomic embeddings** for context-aware search and retrieval.
- **AI-Driven PDF to Structured JSON Extractor**
 - Built end-to-end pipeline to parse PDFs into structured JSON, extracting text, figures, and tables.
 - Trained a custom layout detection model; applied **SmolDocling (VLM)** for layout-aware extraction and **Qwen3** for content categorization.
 - Deployed large models with **vLLM** to support scalable, **high-throughput inference in production.**
 - Enabled **bulk processing** with **Celery + Redis**, later migrated to **Dramatiq + RabbitMQ** for Windows-compatible concurrency and reliability.
 - **Dockerized** and deployed on **Windows (Docker + IIS reverse proxy)** for secure, native rollout.
- **CTS Cheque Extractor & Validator**
 - **Fine-tuned TrOCR** for accurate **handwritten text** extraction from bank cheques.
 - Converted the model to **OpenVINO**, delivering **2–3 times faster throughput** and **reduced CPU inference latency in production.**

- Built a pipeline to extract and validate cheque fields, ensuring high accuracy and efficiency in automated cheque processing.
- **OVD (Officially Valid Document) Masking Services**
 - Built AI-powered masking services with separate **REST APIs** for Aadhaar, PAN, Driving License, and Voter ID.
 - Automatically redacts confidential numbers, QR codes, and barcodes, even in **complex layouts and backgrounds**.
 - **Trained on diverse document samples** to ensure robust detection across varied conditions.
 - Supported multiple formats (**PNG, JPG, PDF, TIFF**), enabling **enterprise integration for privacy and compliance**.
- **Multinational OVD Parsing**
 - Engineered a scalable platform for **parsing information from diverse ID documents** (passports, International cards etc).
 - Designed a pipeline integrating detection, **OCR**, and Named Entity Recognition (NER); **fine-tuned GLiNER** (a compact, bidirectional transformer-based NER model)
 - Implemented **gRPC** for inter-service communication, enabling **low-latency, high-throughput performance** and **efficient handling of multiple model-driven endpoints**.
- **Pothole Detection for Road Safety**
 - Collected and annotated a custom dataset from dashcam footage; trained **YOLOv8** for **real-time pothole detection**.
 - Applied data augmentation to improve robustness across varied road conditions.
 - **Validated the system on live video streams** from an in-car camera setup, demonstrating real-time performance in practical scenarios.
- **Driver Class Prediction**
 - Used data from ECU to create ML models that identifies drivers according to their driving patterns.Used classification models like **KNN,SVM,Decision tree** for model creation

EDUCATION

B.Tech in Computer Science (Artificial Intelligence)

Adi Shankara Institute of Engineering and Technology,Kalady

(2020 - 2024)

SKILLS

- **Programming & Scripting:** Python , Bash
- **Frameworks, Tools & Libraries:** PyTorch, TensorFlow, Hugging Face Transformers, Ollama, FastAPI, OpenVINO, Pandas, NumPy, OpenCV, LangChain, LangGraph, MinIO, K6, Git, Qdrant
- **Machine Learning & AI Techniques:** Machine Learning (ML), Deep Learning (DL), CNNs, OCR, Object Detection, NLP, Generative AI (GenAI), LLMs, VLMs, Retrieval-Augmented Generation (RAG), Agents
- **Data Processing:** Image Processing, Information Retrieval, Document Parsing
- **Deployment & Scaling:** Docker, vLLM, Celery, Dramatiq, Redis, RabbitMQ, gRPC

CERTIFICATION

- **Machine Learning Specialization** - Andrew N G (Coursera)
- **AI Agents in LangGraph** - DeepLearning.AI
- **Prompt Engineering** - DeepLearning.AI